DEPARTMENT OF MATHEMATICS

UNIVERSITY COLLEGE OF ENGINEERING (A):: O.U.
HYDERABAD – 500 007
B.E. – CSE - IV- SEMESTER

CLASS TEST- II – A. Y. 2024 - 25

Engineering Mathematics – IV
(Probability & Statistics)

TIME: 1Hr 30 Min **Max. Marks: 30**

	PART – A (Answer all questions)	5×2 = 10M	Marks	CO	BT
1. a.	Determine the binomial distribution for which mean is 4 and variance is 3. Also find its mode.		2	3	L2
b.	In a Poisson distribution $P(X = 0) = 2P(X = 1)$, then find $P(X = 2)$.		2	3	L2
c.	Write the normal equations to second degree polynomial of the parabola $y = a + bx + cx^2$.		2	3	L2
d.	The coefficient of rank correlation between marks in CA and DBMS by a certain group of students is 0.8. If the sum of the squares of the difference in ranks is given to be 33. Find the number of students in the group.		2	4	L2
e.	Define null hypothesis and level of significance.		2	4	L1
	PART - B (Answer any TWO questions)	**2×10= 20M**			
2. a.	Assume that 50% of all engineering students are good in Mathematics. Determine the probabilities that among 18 engineering students (i) at least 10 (ii) at most 8 (iii) at least 2 and and atmost 9 are good in Mathematics.		5	3	L3
b.	Fit a Poisson distribution to calculate the theoretical frequency for the following data..		5	3	L3

x	0	1	2	3	4
f	109	65	22	3	1

3. a.	In a normal distribution, 7% of the items are under 35 and 89% are under 63. Find the mean and S. D. of the distribution.							5	4	L3
b.	Fit the curve $y = ae^{bx}$ to the following data							5	4	L3

x	1	2	3	4	5	6	7	8
y	15.3	20.5	27.4	36.6	49.1	65.6	87.8	117.6

4. a.	A company manufacturing electric bulbs claims that the average life of its bulb is 16000 hours. The average life and standard deviation of a random sample of 100 such bulbs were 1570 hours and 120 hours respectively. Should we accept the claim of the company?	5	3	L3
b.	A manufacturer claims that only 4% of his products are defective. A random sample of 500 are taken among which 100 defective. Test the hypothesis at0.05 ($Z_{\alpha/2} = 1.96$).	5	4	L3

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